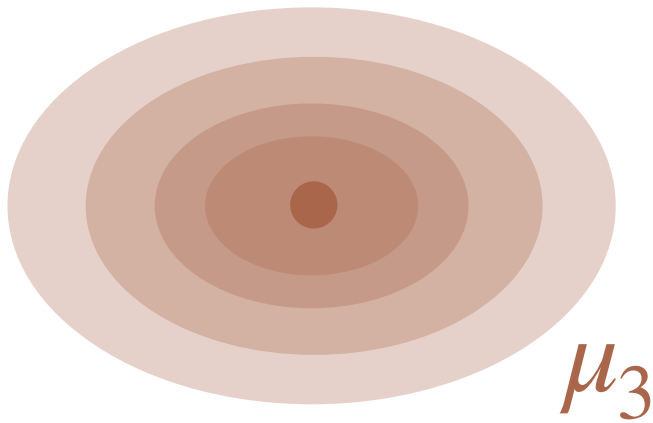
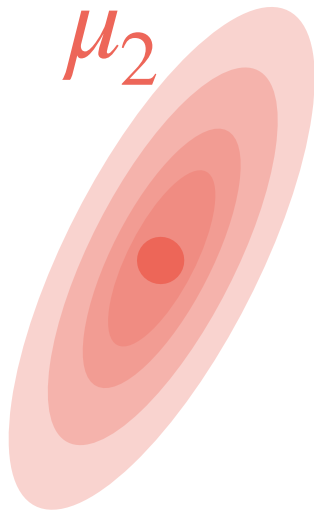




Extends the ideas of means/ends distributions.

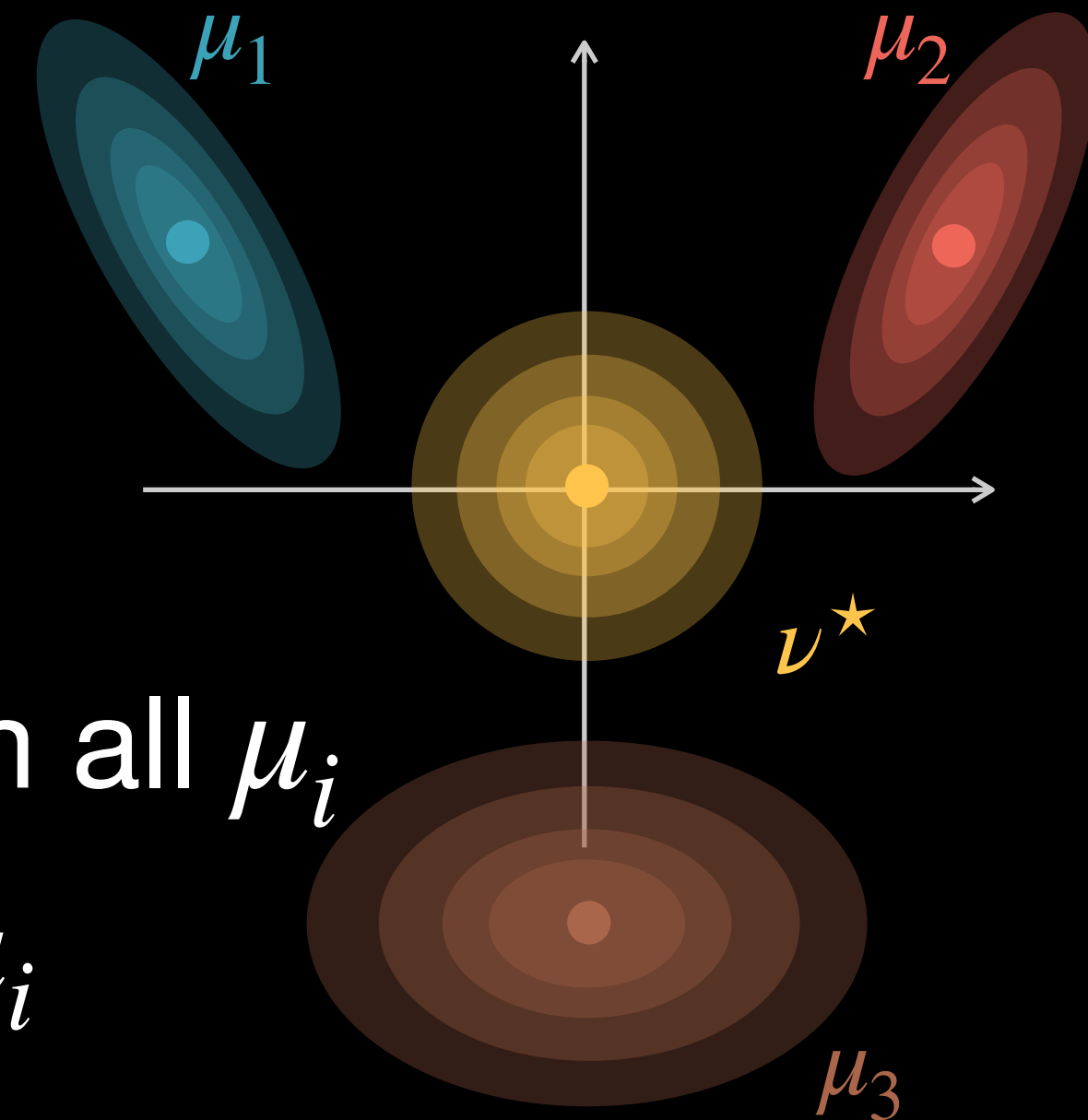
Wasserstein Barycenter

- **Definition:** Given measures $\{\mu_i\}$ and weights $\{\lambda_i\}$ the W_p -barycenter is:

$$\nu^\star = \arg \min_{\nu} \sum_{i=1}^K \lambda_i W_p^p(\mu_i, \nu)$$

- **Interpretation:**

- If all λ_i equal, $\nu^\star$ is distribution that is equidistant from all μ_i
- Else, large $\lambda_i \implies W_p^p(\mu_i, \nu)$ weighs more, $\nu^\star$ closer to μ_i
- A distributional analogue of Fréchet mean ($x^\star = \arg \min_x \sum_{i=1}^K \lambda_i d^p(x, y_i)$)
- **Intuition:** Extends the idea of means/centroids to distributions.



Properties

$$\nu^\star = \arg \min_{\nu} \sum_{i=1}^K \lambda_i W_p^p(\mu_i, \nu)$$